USER GUIDE

Consumer Optimum

Two goods, an income and some prices. Where the consumer stops, why there, and what the 3D surface shows that the plane does not.

BEFORE YOU START

Opening it

There is nothing to install. It is a web page: it opens in Chrome, Safari, Edge or Firefox, on a computer, tablet or phone. You need a connection the first time; after that the browser keeps it and it works offline.

Top right there are two pairs of buttons. The first switches the language between **ES** and **EN**; the second goes from **Dark** to **Light**. For projecting in class, light mode reads better on a beamer; on screen, dark is easier on the eyes.

> **Every graph zooms.** Mouse wheel over the graph, or a two-finger pinch on the trackpad or touchscreen. The zoom applies to the graph under the pointer only, not to the whole page.

> Each slider has a number box beside it. For an exact value, type it into the box and press `Enter`; the slider is for exploring, the box is for precision.

WHAT IS ON SCREEN

The two panels

The screen splits into two views of one problem. They are not two exercises: they are one object seen from two places. Move something in either and the other follows.

PANEL	WHAT IT SHOWS
2D diagram · (x, y) plane	The familiar plane. The background is a heat map of utility — blue low, red high — with the budget line, the indifference curves and the point **P** you can drag.
3D surface · z = u(x, y)	The same function lifted into height. The budget line becomes a vertical plane cutting the surface, and the optimum is the summit of that cut.

> Dragging **P** works in both panels. In the 3D one, dragging on the background — not on the point — rotates the camera.

The readouts

READOUT	HOW TO READ IT
x, y	P's coordinates: how much of each good.
u(P)	Utility at P. It is the height in the 3D panel.
Spend	$p_x \cdot x + p_y \cdot y$. If it is below income, money is left over.
MRS	The marginal rate of substitution at P — the slope of the indifference curve through that point.
$\mathbf{p_x/p_y}$	The relative price — the slope of the budget line.
Gap	The distance between MRS and p_x/p_y. As it approaches zero you are at tangency.
max u	The utility of the true optimum, to compare against yours.

The left-hand panel

CONTROL	WHAT IT DOES
Explore / Challenges	Two modes. In Explore you are in charge: you pick the utility and the prices. In Challenges the applet poses a problem with random numbers and scores your answer.
Preferences	The `u(x, y)` box. Type whatever function you want; it starts at `x^0.5*y^0.5`.
Budget constraint	Three sliders: p_x, p_y and income m. They start at 2, 3 and 24.
Graphical method	How you hunt for the optimum: *Pick the bundle* (drag P along the line), *Raise the level curve* (raise u until it touches) or *Free point* (move P anywhere in the quadrant).
Layers	Turns each element of the drawing on and off. Fewer layers is nearly always better for explaining; switch them on one at a time.
Domain	How far the axes reach. Raise it if the optimum leaves the frame.
Fit to budget	Puts P back on the budget line if you have taken it off.

The layers, one by one

LAYER	WHAT IT DRAWS
Utility map	The coloured background. Blue is low utility, red is high.
Contours	Indifference curves at regular levels, like a topographic map.
Level curve	The curve at the level you set with the slider.
Curve through P	The indifference curve running exactly through P.
Feasible set	What you can afford: the triangle under the line.
3D slice	In the 3D panel, the curve where the budget plane cuts the surface.
Tangent at P	The tangent to the indifference curve at P. When it lines up with the budget line, you are there.
Gradient	The vector ∇u at P: it points where utility grows fastest.
Partials	The two components of the gradient separately, u_x and u_y.
Cutting plane	The vertical budget plane, in the 3D panel.
Affordable only	Clips the 3D surface to the feasible set: the rest disappears.
Show optimum	Marks the true solution. Turn it off while the student searches.
Leave a trail	P leaves a trail of marks as it moves.

THE FIRST TEN MINUTES

A guided run

Open it and do this in order. The results are checked against the starting values, so if they do not match something has moved: reload the page and start again.

1. Leave it as it comes: `u = x^0.5*y^0.5`, p_x = 2, p_y = 3, m = 24. Switch on **Show optimum**.

   ```
   The optimum sits at x* = 6, y* = 4, with u = 4.899.
   ```

2 Check the arithmetic by hand: with equal Cobb–Douglas exponents each good takes half the income. 12 on x at 2 each is 6 units; 12 on y at 3 each is 4.

```
Spend = 2·6 + 3·4 = 24. Nothing left over.
```

3 Switch on **Tangent at P** and drag P along the line until the tangent lies flat on the line itself. Watch the **Gap** as you do it.

> At the optimum, MRS = 0.667 = p_x/p_y = 2/3, and the gap is zero.

4 Switch the method to **Raise the level curve** and raise u slowly. You will watch the curve go from cutting the line twice, to grazing it, to missing it.

```
The level where it stops cutting is exactly u = 4.899.
```

5 Now look at the 3D panel. The vertical plane cuts the surface in an arch; the optimum is its highest point. Rotate the camera by dragging the background.

```
Both panels mark the same point at the same time.
```

THREE EXPERIMENTS

1 · When tangency is no use

The condition MRS = p_x/p_y is what gets taught, but it only holds for interior solutions with smooth curves. The next two cases break it, and the applet says so.

1 Type `min(x,y)` and leave p_x = 2, p_y = 3, m = 24. The indifference curves are right angles.

```
x* = y* = 4.8. The applet calls it a “kink”: MRS does not
exist there.
```

2 Now type `2x+y`, perfect substitutes. MRS is 2 everywhere; the relative price is 2/3.

```
x* = 12, y* = 0. A corner: MRS never gets down to the price
ratio.
```

3 With the same substitutes, raise p_x past 6 (that is, until p_x/p_y goes above 2).

```
The solution jumps straight to the other axis: everything in
y. There is no smooth transition.
```

2 · The gradient, and why it points outward

Switch on **Gradient** and **Partials** and drag P along the budget line. The gradient always points uphill on the surface; at the optimum it ends up perpendicular to the line. That is the same tangency condition, said with vectors.

3 · The challenges

Press **Challenges**. There are six levels and each one generates fresh numbers every time, so the answer cannot be memorised. They unlock in order.

LEVEL	WHAT IT DRILLS
1 · Cobb–Douglas	Interior tangency with equal exponents. The textbook case.
2 · Asymmetric Cobb–Douglas	The same with unequal exponents, by the level-curve method.
3 · Perfect substitutes	Corner solutions and the jump from one axis to the other.
4 · Perfect complements	The kink, where MRS is undefined.
5 · Quasilinear	One good's demand does not depend on income; the other absorbs the rest.
6 · CES	Elasticity of substitution away from one, with ρ positive or negative.

> The score rewards getting close to the true optimum. **Check** grades it, **Try again** repeats with the same numbers and **Next** moves on.

WRITING YOUR OWN FUNCTION

The syntax it accepts

In the `u(x, y)` box the variables are **x** and **y**, and only those. The partial derivatives are computed exactly, not approximated, so the MRS you see is the real one.

CONTROL	WHAT IT DOES
`+ − * / ^`	Add, subtract, multiply, divide, power. `x^0.5` is the square root of x.
`( )`	Brackets. `(x+y)^2` is not the same as `x+y^2`.
`sqrt ln log exp abs`	Root, natural log, base-10 log, exponential and absolute value.
`min max`	Two arguments: `min(x,y)` is perfect complements.
`sin cos`	Trigonometric, in radians. Rarely useful here, but available.
`e pi`	The two constants. `e` is 2.718…, `pi` is 3.1416…
`2x`, `3xy`	Implicit multiplication works: `2x` reads as `2*x`.

Examples that work

EXPRESSION	WHAT PREFERENCES IT DESCRIBES
`x^0.5*y^0.5`	Symmetric Cobb–Douglas.
`x^0.3*y^0.7`	Cobb–Douglas weighted towards y.
`min(2x,y)`	Fixed-proportion complements: two x per y.
`3x+y`	Perfect substitutes with a constant MRS of 3.
`4ln(x)+y`	Quasilinear. The optimum in x does not depend on income: $x^* = 4p_y/p_x$.
`(x^0.5+y^0.5)^2`	CES with $\rho = 0.5$, more substitutable than Cobb–Douglas.
`(x^(-1)+y^(-1))^(-1)`	CES with $\rho = -1$, less substitutable.

> If it cannot be read you get *"I can't read that expression"* and the graph stays as it was. The usual causes are an unclosed bracket, a decimal comma instead of a point (`0,5` instead of `0.5`) or a variable other than x and y.

IF SOMETHING GOES WRONG

Common problems

SYMPTOM	WHAT TO DO
The optimum leaves the frame	Raise the **Domain**, or lower income. With large m and small prices the optimal bundle runs far out.
The 3D surface looks flat	You are looking at it almost edge-on. Drag the panel background to rotate the camera and raise the elevation.
I cannot drag P	In the 3D panel you must grab the point, not the background: the background rotates the camera. If P has left the line, press **Fit to budget**.
The indifference curve does not appear	With functions like `ln(x)+ln(y)` utility is negative over part of the quadrant and there is no curve to draw there. Try `x^0.5*y^0.5`.
The challenges are locked	They unlock in order: you must clear the previous one. Press **Check** with P at the optimum.

Open the applet: Consumer Optimum. It all runs in the browser, and once loaded, offline too.

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